



**IRAMCO INSTITUTE OF TECHNOLOGY
RAJAPALAYAM, TAMILNADU, INDIA**

**DEPT. OF COMPUTER SCIENCE AND ENGINEERING
ACADEMIC YEAR – 2019-2020 (EVEN)**

Course code & Title: CS8603-DISTRIBUTED SYSTEMS

Semester - VI Date:07.02.2020 Time:09.50 am to 10.40 am

**Topic: Distributed Mutual Exclusion-
Introduction**

FLIPPED CLASS COLLABORATIVE LEARNING

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1. Objectives

- To enhance communication, listening, Self-confidence and problem solving skills
- To describe the distributed mutual exclusion.

2. Outcome (O) (At the end of the learning technique, the students will be able to)

- Describe the distributed mutual exclusion and it's needs.

3. Justification for choosing the topic

Nowadays, computer systems are presented in almost all types of human activity and they support any kind of industry as well. Most of these systems are distributed where the communication between nodes is based on computer networks of any kind. Connectivity between system components is the key issue when designing distributed systems, especially systems of industrial informatics for sharing resources. For example, data produced in one physical location and required in another location. When multiple systems try to access the shared resources, the distributed system must ensure that only one system is allowed to access the shared resources at any given instance of time. It can be achieved by distributed mutual exclusion technique. Based on this, most of the industrial application viz WWW, telephone networks, cellular networks, massively multiplayer online games, VR communities and etc are working. The objective of collaborative learning is to enhance communication, creativity and problem-solving skills. This domain helps the students to get knowledge in the working of telephone networks, cellular networks, WWW and etc.

A collaborative learning technique highlights the students with communication and improves their problem-solving skills through the series of assessments. Because of most of the students like autonomy and want to learn by their own, I consider this topic for self-learning.

4. Flipped classroom

Flipped classroom is a pedagogical approach [1] in which traditional content delivery moves to an *individual self learning process*, and the group activity is transformed into a dynamic, interactive learning environment where the instructor just guides the learners to apply their creative thinking to provide a fruitful solution in the course content. It provides more benefits than traditional direct content delivery.

4.1. Benefits of flipped classroom

- It encourage peer interaction among students and *collaborative learning skills*
- It increases individual attention.
- It makes *self-learning* rather than teaching.

4.2. Procedure:

Plan

- Identify the topic for *self study*.
- Choose the date and time for flipped class which contains date for posting video material, class activity.

Prepare

- Identify existing recorded video from MOOCs (NPTEL/MIT etc.) and text materials or create a video lecture
- Write the guidelines for learning from the video

Implement

- Upload the learning materials such as video and text documents on the course website (Canvas)
- Give minimum 3 to 5 days for students to watch the video.
- Instruct the students to watch the video individually from their home and take the notes according to their understanding level.
- On the day of the activity, form a small group. A group may be heterogeneous or homogeneous based on the students' performance. The fig 1 shows the various group with members.



Fig.1. Various groups with peer members

- Each group members discuss among the peer members based on their self-learning and write the points on paper for panel discussion (Presentation) within 10 minutes. The fig 2 and 3 shows the group members discussions and content written on the paper.



Fig.2. Group members write their points on paper

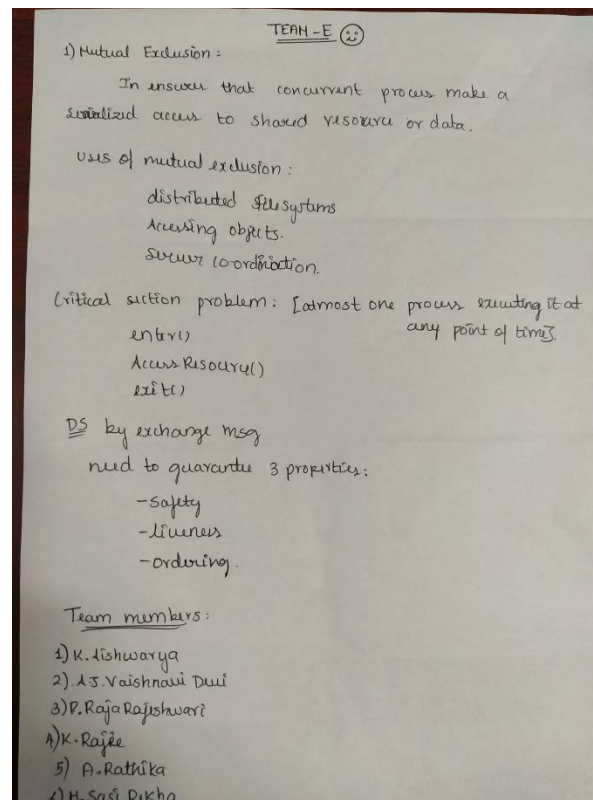


Fig.3. Sample Important Points discussion about Mutual Exclusion by Team E

- Once they prepared the content for preparation after discussion with their peer members, another 5 minutes are given to each group for presentation. The fig.4 shows the sample group presentation.



Fig.4.Team 'A' presents their Views

4.3. Ground Rule

- Learning materials such as NPTEL Video on Distributed Mutual exclusion and its need are uploaded on the course website (Canvas).
- Open to suggestions, sensitive feelings and concerns
- Groups are not allowed to repeat same points about the topic which were discussed already by other group.

5. Reflection critique

Challenges

- Recording video lecture and delivering content in front of the camera is challenging.
- Recording Quality video and getting the quality device.
- Deadlock situation may happen during an in-class activity like brainstorming

Steps to be taken to avoid the problem.

- After recording a video with the help of my colleague, I gave it to peer-review to get both positive and negative suggestions/feedback from my peers. Based on their suggestion, I prepared a quality short video lecture.
- The Presentation (PPT) is integrated with video.
- Tips were given to motivate the students during the participation of in-class activity

Post Implementation

1. They observed and learned a few things individually that lead to self-learning. They *actively involved* in this session rather than it being a one-way communication.
2. All the groups performed well in the group presentation session. This session *re-enforced their understanding*.
3. One of the Group (named as Group A), all the members involved to present their own views rather than other groups.
4. Slow learners from two groups were not comfortable in their presentation due to lack of attention. However, they understood the concepts by these learning methods.
5. Learning at home automatically becomes *student-centric* rather than teacher-centric

Relevance of program outcomes

Programme Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
Outcome(O)	3	2	2	2	-	-	-	2	-	2	-	2

Conclusions

The selected topic is playing a vital role in the industry to develop distributed applications for sharing the resources. The flipped classroom technique is used to assess the students' performance in continuous progress. The students could be able to enrich various skills such as *communication, cooperation, and leadership*.

References:

1. Flipped Learning Network (FLN). (2014) The Four Pillars of F-L-I-P™
2. Stephanie butler velegol, sarah e. Zappe, emily mahoney, The Evolution of a Flipped Classroom: Evidence-Based Recommendations, winter 2015, pp.1-37
3. Dian Mita Nurhayati and Hartono, Implementation of Cooperative Learning Model Type with RME Approach to Understanding of Mathematical Concept Student State Junior High School in Pekanbaru, AIP Conference Proceedings 1848, 040002, 2017, <https://doi.org/10.1063/1.4983940>
4. Micheal M van Wyk, The Effects of the Cooperative Learning Method on Student Achievement, Attitude and Motivation in Economics Education, Journal of Social Science pp. 261-270, 2017, <https://doi.org/10.1080/09718923.2012.11893104>

5. Dr.M.Kaliappan, “Web Services and its applications” in course “IT6801 Service Oriented Architecture” dated:25.10.2019.
6. Mr.M.Gomathy Nayagam, “HDFS Concepts” in course “CS6703 Grid and Cloud Computing” dated:16.09.2019.

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